PLPBP-Related Pyridoxine-Dependent Epilepsy (PLPHP Deficiency): Comprehensive Disease Characterization

Target disease: PLPBP-Related Pyridoxine-Dependent Epilepsy (Developmental and Epileptic Encephalopathy 2024; vitamin B6-dependent epilepsy due to PLPHP deficiency) **MONDO:** MONDO:0015005 (pyridoxine-dependent epilepsy, umbrella). **Category:** Mendelian, autosomal recessive. **Evidence base:** 27 literature sources reviewed; 11 confirmed findings recorded. This is a literature-based synthesis (no primary patient dataset provided). Evidence types are flagged as *human clinical*, *model organism*, *in vitro/biochemical*, or *computational/structural*.

> **Important caveat on rarity:** As of 2026 only ~54 patients have been reported worldwide (Alsini et al. 2026, P 41671913).
> Many disease parameters (precise prevalence, penetrance, QoL metrics, survival curves) are therefore **not formally established** and are noted as such.

---

1. Disease Information

PLPBP-related pyridoxine-dependent epilepsy is a rare autosomal-recessive inborn error of **vitamin B6 (pyridoxal 5'-phosphate, PLP) homeostasis** that presents as a **neonatal-onset developmental and epileptic encephalopathy**. Seizures are typically refractory to conventional antiepileptic drugs (AEDs) but respond to vitamin B6 (pyridoxine or PLP). It was first defined molecularly in 2016 when biallelic *PROSC* (now *PLPBP*) mutations were shown to "disrupt cellular pyridoxal phosphate homeostasis and cause vitamin-B6-dependent epilepsy" (Darin et al.,

).

P 27912044

Information source: aggregated individual case reports and small cohorts/systematic reviews (not population EHR).

Key identifiers - **Gene:** *PLPBP* (formerly *PROSC*); HGNC:30042; NCBI Gene 65015; Ensembl ENSG00000147471; UniProt O94903 (PLPHP_HUMAN). Locus **8p11.23**. - **Protein:** PLPHP / PLPBP (pyridoxal 5'-phosphate homeostasis protein). - **OMIM:** gene *PLPBP* #604436; phenotype **#617290** (Developmental and epileptic encephalopathy 2024, formerly EIEE/pyridoxine-dependent). - **Orphanet:** within "Pyridoxine-dependent epilepsy" (ORPHA:3006 umbrella; PLPHP deficiency as a molecular subtype). - **ICD-11:** 8A61 (developmental and epileptic encephalopathies) / 5C50.A (disorders of vitamin B6 metabolism). **ICD-10:** G40.4 + E53.1 (vitamin B6 deficiency) as proxies. - **MeSH:** "Seizures" / "Epilepsy" + "Vitamin B 6"; no dedicated MeSH heading for PLPHP deficiency. - **MONDO:** MONDO:0015005 (umbrella, pyridoxine-dependent epilepsy).

Synonyms / alternative names: PLPHP deficiency; PLPBP deficiency; PROSC-related/PROSC deficiency epilepsy; Vitamin B6-dependent epilepsy (PLPBP type); Pyridoxine-dependent epilepsy type 2/PLPBP; Developmental and epileptic encephalopathy 2024.

---

2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound heterozygous) pathogenic variants in *PLPBP*. "*Biallelic pathogenic variants in PLPBP (formerly called PROSC) have recently been shown to cause a novel form of vitamin B6-dependent epilepsy*" (Johnstone et al. 2019,

).

P 30668673

No environmental or infectious primary cause; the disease is monogenic.

Genetic risk factors. The disease-causing variants themselves are the risk factor. **Consanguinity** strongly elevates risk (many reported families are consanguineous; e.g., a first-cousin Pakistani family, Ahmed et al. 2020,

).

P 33425341

A recurrent/founder deletion **c.370_373del (p.Asp124fs)** is the single most frequent variant (18.5% of alleles; Alsini 2026,

; P 41671913 also a founder allele reported by Pal et al. 2021,). P 33977028

Environmental risk factors. None established as causal. However, **intercurrent viral illness/fever can trigger breakthrough seizures or metabolic crises** (Takle et al. 2025,) P 42563912 — a gene–environment interaction where physiological stress increases demand on an already-compromised PLP-homeostasis system. **Age:** risk window is prenatal/neonatal (congenital metabolic demand). **Sex:** no predilection (AR).

Protective factors. No validated genetic protective alleles or modifier genes are published. The principal *modifiable* protective factor is **early vitamin B6 supplementation**, which is treatment rather than a constitutional protective factor. Residual protein function (hypomorphic missense vs. null) is inferred to be protective against the most severe phenotype (genotype–severity correlation, Finding 10).

Gene–environment interaction. Illness-provoked crises (above) are the clearest example; otherwise GxE data are *not available* for this ultra-rare disease.

---

3. Phenotypes

Core phenotype = **neonatal developmental and epileptic encephalopathy**. Data below are from case series and the 54-patient systematic review (Alsini 2026, ; P 41671913 Johnstone 2019,). P 30668673

Phenotype	Type	Onset	Severity	Frequency	HPO
Neonatal seizures (multifocal, clonic, tonic, myoclonic, status epilepticus)	Clinical sign	Neonatal (55.5% <24 h; 76% <1 week)	Severe	~100%	HP:0032807 (Neonatal seizure), HP:0002133 (Status epilepticus), HP:0032794 (Clonic seizure)
AED-resistant, B6-responsive seizures	Clinical sign	Neonatal	Severe	~100%	HP:0011097 (Epileptic spasm), HP:0001250 (Seizure)
Global developmental delay / intellectual disability	Clinical sign	Infancy onward	Moderate–severe (>2/3 patients)	Majority	HP:0001263, HP:0001249
Microcephaly (often congenital)	Physical manifestation	Congenital/ early	Variable	Frequent	HP:0000252
Encephalopathy	Clinical sign	Neonatal	Severe (untreated)	Common	HP:0001298
Movement disorder: myoclonus, tremor, abnormal eye movements (OMAS-like)	Clinical sign	Infancy/ childhood	Variable, episodic	Subset	HP:0001336 (Myoclonus), HP:0002072, HP:0000496
Infantile spasms (± hypsarrhythmia)	Clinical sign	Infancy	Variable	Subset	HP:0012469
Prematurity, fetal distress, antenatal anomalies	Perinatal	Prenatal/ birth	—	Common	HP:0001622, HP:0001623
		Neonatal	—	Subset	HP:0003128

Phenotype	Type	Onset	Severity	Frequency	HPO
Lactic acidemia (nonspecific)	Lab abnormality				
Hyperglycinemia / elevated glycine & glutamine	Lab abnormality	Neonatal	—	Subset	HP:0002154 (Hyperglycinemia)

Progression: Seizures are **episodic/fluctuating** acutely and become **treatment-controlled** on B6; the underlying encephalopathy is largely **static** but with persistent developmental impairment. **Quality of life:** no formal EQ-5D/PROMIS data exist; QoL is reduced in proportion to intellectual disability, epilepsy burden, and microcephaly (*not formally quantified*).

---

4. Genetic / Molecular Information

- **Causal gene:** *PLPBP* (HGNC:30042; OMIM 604436; 8p11.23). Encodes PLPHP (296 aa; UniProt O94903).
- **Variant spectrum (Alsini 2026,**

  P 41671913

  ;
  Johnstone 2019,

  P 30668673

  ):
  missense ≈48% (most common class), plus frameshift, nonsense, splice-site, and small indels. Representative variants: **c.370_373del (p.Asp124fs)** — most frequent/founder (18.5%); **c.46_47insCA (p.Leu17Hisfs)** (Ahmed 2020,

  P 33425341

  );
  c.695C>T (p.Ala232Val) missense (İpek 2022,

  P 36308585

).
- **Classification (ACMG/AMP):** reported variants are Pathogenic/Likely pathogenic in ClinVar; biallelic requirement.
- **Allele frequency:** individually ultra-rare in gnomAD (consistent with AR disease; carrier frequency not precisely established, inferred very low outside founder contexts).

- **Origin: germline** (constitutional). No somatic involvement.
- **Functional consequence:** predominantly **loss of function** — null alleles or missense that reduce protein **stability and/or PLP-binding**. "The most severe phenotypes were associated with variants leading to loss of function of PLPBP or significantly affecting protein stability/ PLP-binding" (Johnstone 2019,

P 30668673

).
- **Modifier genes / epigenetics / chromosomal abnormalities:** none established; no reported CNVs, methylation signatures, or cytogenetic lesions. *Not applicable / not available.*

---

5. Environmental Information

- **Environmental factors / toxins / occupational exposures:** none causal (*not applicable*).
- **Lifestyle factors:** not relevant to a neonatal monogenic disorder; however **dietary vitamin B6 status** modulates the biochemical state, and B6 supplementation is therapeutic.
- **Infectious agents:** none cause the disease, but **viral illnesses act as triggers** for seizure/ metabolic decompensation (Takle 2025,

P 42563912

).
NCBI Taxonomy not applicable to etiology.

---

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function *PLPBP* variants** → reduced amount or stability/activity of **PLPHP** protein (*demonstrated*: protein-stability/PLP-binding modeling, Johnstone 2019

P 30668673

).
2. Loss of PLPHP → **failure of intracellular PLP homeostasis/chaperoning** — the cell can no longer safely buffer and deliver the reactive aldehyde PLP to apo-enzymes (*demonstrated in vitro/structurally*: He 2022

;
P 35897955
"mediates PLP homeostasis by delivering PLP to target enzymes or by performing a protective function").

3. This **leads to** two branches:
4. **Branch A (substrate starvation):** many **PLP-dependent enzymes** receive insufficient correctly-delivered PLP → **reduced holoenzyme activity**.
5. **Branch B (toxic vitamer accumulation):** aberrant accumulation of **pyridoxine 5'-phosphate (PNP)** (*demonstrated in bacteria/in vitro*: Ito 2020
;
P 31677193
Ito 2022
),
P 34978460
which **directly inhibits** PLP-dependent enzymes (e.g., the **glycine cleavage system**).
6. Both branches **result in** dysregulation of **amino-acid and neurotransmitter metabolism** — notably **GABA synthesis (glutamate decarboxylase, GAD)**, **glycine catabolism**, and **catecholamine/monoamine synthesis (aromatic L-amino-acid decarboxylase, AADC)** (*model-organism evidence*: zebrafish *plpbp*−/− shows "disruptions in amino acid metabolism as well as GABA and catecholamine pathways", Johnstone 2019
).
P 30668673
7. Imbalanced excitatory/inhibitory neurotransmission (inferred: ↓GABA, altered glycine/glutamate) → **neuronal hyperexcitability → refractory neonatal seizures** (*phenotype demonstrated*; precise neurotransmitter stoichiometry *inferred*).
8. Sustained seizures + metabolic derangement during a critical developmental window → **encephalopathy, white-matter injury, impaired brain growth (microcephaly) → developmental delay / intellectual disability**; untreated → **status epilepticus and death in infancy** (Johnstone 2019
).
P 30668673
9. **Therapeutic branch:** exogenous **vitamin B6 (pyridoxine → PLP, or direct PLP)** raises PLP availability, partially restoring enzyme function → **seizure control** (*demonstrated* in humans and rescued in zebrafish,
).
P 30668673

Supporting detail by category

- **Molecular pathways / metabolic changes:** vitamin B6 (PLP) metabolism and dependent pathways — amino-acid metabolism (glycine cleavage, aspartate/serine), neurotransmitter biosynthesis (GABA via GAD, dopamine/serotonin via AADC), transsulfuration. KEGG: vitamin B6 metabolism (map00750). GO: vitamin B6 metabolic process (GO:0042816), pyridoxal phosphate biosynthetic process (GO:0042823), neurotransmitter biosynthetic process (GO:0042136).
- **Protein dysfunction:** PLPHP has a **type III TIM-barrel fold** (He 2022, P 35897955) with a conserved phosphate-coordinating PLP pocket; pathogenic missense destabilize the fold or the PLP site (loss of function; no dominant-negative or gain-of-function reported). PDB homologs exist for bacterial YggS; human structure via AlphaFold (O94903).
- **Biochemical abnormalities:** functional insufficiency of multiple PLP-dependent enzymes; accumulation of PNP; nonspecific hyperglycinemia and lactic acidemia (Jensen 2019, P 31741821).
- **Moonlighting / non-canonical functions:** PLPHP is **also an RNA-binding protein** (Graziani 2024, P 39604152) and has a **mitochondrial pool** linked to proteostasis (accumulates when mitochondrial peptidase CLPP is lost; Key 2023, P 38139332); PLP-independent functions are reported for the cyanobacterial ortholog PipY (Llop 2026, P 41820517). These broaden the mechanism beyond cytosolic PLP delivery and may explain **mitochondrial-disease mimicry** and nonspecific lactic acidemia.
- **Cellular processes / tissue damage:** neuronal hyperexcitability and excitotoxic/seizure-related injury (inferred); no primary inflammatory or autoimmune component. **Immune involvement: not applicable.**
- **Cell types / GO-CC:** neurons (CL:0000540), especially **GABAergic neurons** (CL:0000617); glia secondary. Subcellular: **cytosol (GO:0005829)** and **mitochondrion (GO:0005739)**;

molecular function **pyridoxal phosphate binding (GO:0030170)**, **RNA binding (GO:0003723)**.

- **CHEBI chemical entities:** pyridoxal 5'-phosphate (CHEBI:18405), pyridoxine (CHEBI:16709), pyridoxine 5'-phosphate (CHEBI:58589), γ-aminobutyric acid (CHEBI:16865), glycine (CHEBI:15428).
- **Molecular profiling (omics):** no human transcriptomic/proteomic/metabolomic disease atlas published; metabolic perturbations characterized in zebrafish and bacterial models (*model-organism evidence*).

---

7. Anatomical Structures Affected

- **Organ/system: central nervous system / nervous system (UBERON:0001016)** is the primary target; brain (UBERON:0000955). No consistent extra-CNS organ disease.
- **Regional (imaging, Alsini 2026,**

  ):

  P 41671913

  abnormalities in **65%** — **white matter (UBERON:0002316) abnormalities 54%**, **periventricular/temporal cysts 27%**, **corpus callosum (UBERON:0002336) anomalies 15.4%**, global brain underdevelopment with broad gyri/shallow sulci; cerebral cortex (UBERON:0000956) involvement.
- **Tissue/cell:** nervous tissue; neurons (CL:0000540), GABAergic neurons (CL:0000617); glia/white-matter tracts secondarily.
- **Subcellular:** cytosol (GO:0005829) and mitochondrion (GO:0005739).
- **Lateralization: bilateral**, diffuse/global (consistent with a metabolic encephalopathy); not focal/lateralized.

---

8. Temporal Development

- **Onset: congenital/neonatal.** Seizures begin very early — 55.5% within the first 24 h and 76% within the first week of life (Alsini 2026,

  );

  P 41671913

rare later-onset (infantile) cases exist. Onset pattern: **acute** seizures on a congenital metabolic background. Some fetal seizure activity/antenatal anomalies reported.

- **Progression:** acute refractory phase → **treatment-induced remission** of seizures on B6 → **chronic, largely static encephalopathy** with persistent developmental impairment. Disease course for seizures is **episodic** (with illness-triggered breakthroughs) superimposed on a stable deficit.
- **Duration: chronic, lifelong**; requires lifelong B6 supplementation (withdrawal → seizure recurrence).
- **Remission: treatment-induced** (vitamin B6), not spontaneous.
- **Critical period:** the **neonatal/early-infantile window** is the key opportunity for intervention; earlier treatment limits (but does not eliminate) neurodevelopmental damage.

---

9. Inheritance and Population

- **Inheritance: autosomal recessive.** Biallelic variants required; parents are obligate carriers.
- **Epidemiology: ultra-rare.** Overall pyridoxine-dependent epilepsy birth incidence ≈ **1:396,000 (definite/probable) to 1:252,000 (incl. possible)** in the Netherlands (Been 2005, P 16159904) — figures dominated by ALDH7A1. **PLPBP is a small minority**: ~30 patients by 2020 (Ahmed, P 33425341), **54 by 2026** (Alsini, P 41671913). Point prevalence: not formally established (<1/1,000,000, inferred).
- **Penetrance:** appears **complete** for biallelic LoF (all reported biallelic patients are affected); formal penetrance estimates unavailable.
- **Expressivity: variable** — severity correlates with residual protein function (LoF/unstable = severe).
- **Anticipation: not applicable** (not a repeat-expansion disorder).
- **Germline mosaicism:** not reported.

- **Founder effects: c.370_373del** acts as a recurrent/founder allele (Pal 2021, P 33977028; Alsini 2026).
- **Consanguinity:** major contributor; many homozygous cases in consanguineous families.
- **Carrier frequency:** very low, not precisely quantified.
- **Demographics:** reported worldwide (Europe, Middle East/Saudi Arabia, Pakistan, North America, East Asia); **higher counts in consanguineous populations**. **Sex ratio ≈ 1:1** (AR). **Age distribution:** overwhelmingly neonates/infants at presentation.

---

10. Diagnostics

- **No specific diagnostic metabolite** (unlike ALDH7A1 → ↑α-AASA/pipecolic acid, or PNPO). Biochemistry is **nonspecific**: "diagnostic pitfalls, including lactic acidemia as well as hyperglycinemia, highlighting the importance of a pyridoxine trial, and genetic testing" (Jensen 2019, P 31741821); elevated glycine/glutamine may occur (Takle 2025, P 42563912). LOINC-codable labs: plasma amino acids (glycine), lactate, CSF neurotransmitter metabolites.
- **Clinical/functional tests: EEG** (multifocal/burst-suppression or spasms; infantile spasms sometimes without hypsarrhythmia — Kalser 2023, P 36577449); **therapeutic trial of pyridoxine (± PLP)** with seizure response is a key diagnostic maneuver.
- **Imaging:** brain **MRI** frequently abnormal (65%) — white-matter changes, periventricular/temporal cysts, callosal anomalies, atrophy (Alsini 2026, P 41671913). RadLex: cerebral white matter lesion.

- **Genetic testing (definitive): WES or WGS**, or **epilepsy/vitamin-B6 gene panels** including *PLPBP*; single-gene *PLPBP* sequencing when clinically suspected. WES often succeeds after nondiagnostic panels (Takle 2025,

  P 42563912

). CMA/karyotype/FISH generally unrevealing (no CNVs reported). Mitochondrial DNA/repeat-expansion testing not indicated (though mitochondrial disease is a common misdiagnosis).
- **Clinical criteria & differential diagnosis:** diagnosis rests on **B6-responsive neonatal epilepsy + biallelic *PLPBP* variants**. Differentials: **ALDH7A1** (classic PDE; ↑α-AASA), **PNPO** deficiency (PLP-responsive), **hypoxic-ischemic encephalopathy**, **mitochondrial encephalopathy**, **folinic-acid-responsive seizures** (allelic with ALDH7A1), **AADC deficiency**, **nonketotic hyperglycinemia**, and **opsoclonus-myoclonus-ataxia syndrome (OMAS)** (Johnstone 2019

  P 30668673

  ; Takle 2025

  P 42563912

).
- **Screening: not on standard newborn screening** (no specific metabolite). **Cascade/carrier testing** in families with a known variant and **prenatal/preimplantation testing** are feasible once the familial variants are known.

11. Outcome / Prognosis

- **Survival/mortality: Untreated disease is fatal** — "can progress to status epilepticus and death in infancy" (Johnstone 2019,

  P 30668673

). With treatment, survival into childhood/adulthood is expected; formal survival rates unpublished.
- **Morbidity/function: Guarded** — **>2/3 of patients classified moderate-to-severe** on a combined neurodevelopmental/seizure-control severity score (Alsini 2026,

  P 41671913

).

Common sequelae: intellectual disability, developmental delay, microcephaly, ongoing epilepsy risk.

- **Disease course/complications:** chronic static encephalopathy; breakthrough seizures during illness; feeding/motor impairment in severe cases.
- **Recovery potential:** seizures are usually controllable with B6, but **neurocognitive recovery is incomplete**; earlier treatment improves but does not normalize outcomes.
- **Prognostic factors: variant class (LoF/unstable → worse)**, treatment delay, neonatal seizure burden, degree of MRI abnormality/microcephaly. **Prognostic biomarkers:** none validated; genotype is the best available predictor. QoL: no EQ-5D/SF-36/PROMIS data (*not available*).

12. Treatment

- **Pharmacotherapy (disease-specific): Vitamin B6.** "neonatal onset seizures responsive to treatment with vitamin B6 available as **pyridoxine (PN)** or as the biologically active form **pyridoxal 5-phosphate (PLP)**" (Ahmed 2020, P 33425341).
  Seizures are "generally resistant to traditional antiepileptic drugs and show a dramatic response to high-dose pyridoxine" (İpek 2022, P 36308585).
  Some patients respond to pyridoxine, others require PLP; **lifelong supplementation** is mandatory (withdrawal → recurrence). NCIT: **Pyridoxine (C745)**, **Pyridoxal Phosphate (C84305)**, Pyridoxine Therapy. CHEBI: pyridoxine (CHEBI:16709), PLP (CHEBI:18405). DrugBank: Pyridoxine DB00165; Pyridoxal phosphate DB00114.
- **Pharmacogenomics / personalized:** treatment is **genotype-directed** — confirming *PLPBP* distinguishes it from ALDH7A1/PNPO and dictates B6 (not diet) therapy. PN vs PLP choice is individualized.
- **Supportive AEDs:** conventional AEDs are largely ineffective as monotherapy but may be adjuncts; standard supportive/rehabilitative care (PT/OT/speech), developmental support, nutrition.
- **Therapies that do NOT apply: lysine-reduction therapy** (dietary lysine restriction ± arginine), which benefits ALDH7A1 PDE by lowering α-AASA (Coughlin 2022,

), P 36008148 is **mechanistically irrelevant** to PLPBP (no α-AASA accumulation).

- **Advanced therapeutics: No approved gene, cell, or RNA therapy.** Gene replacement is conceptually attractive (monogenic LoF) but **not in clinical trials** for *PLPBP* (*not available*). No disease-specific NCT trials identified.
- **Treatment outcomes / adverse events:** B6 generally controls seizures; **high-dose pyridoxine carries a risk of sensory peripheral neuropathy** (a general class effect) — dosing should balance efficacy and toxicity (monitor). FAERS: pyridoxine neuropathy at high chronic doses.

13. Prevention

- **Primary prevention:** not possible at the individual level (monogenic, congenital). **Genetic counseling** for carrier couples (25% recurrence risk per pregnancy, AR) and **reproductive options** (prenatal diagnosis, PGT-M) where the familial variants are known.
- **Secondary prevention: early recognition and prompt B6 trial** in any neonate with unexplained AED-resistant seizures; early molecular diagnosis (WES/WGS) to start targeted therapy within the critical window.
- **Tertiary prevention:** lifelong B6 to prevent seizure recurrence; **supplement during intercurrent illness** to avert illness-triggered crises; developmental/rehabilitative support to mitigate disability.
- **Immunization / public-health / environmental measures:** not applicable (no infectious or environmental cause), though routine childhood vaccination to reduce febrile-illness triggers is prudent supportive advice.
- **Counseling:** AR genetic counseling is the cornerstone of prevention (NSGC/ACMG frameworks); cascade carrier testing for relatives.

14. Other Species / Natural Disease

- **Evolutionary conservation:** the **YggS/Ybl036c/PLPBP (COG0325) family is "widely conserved across the three domains of life"** (Xu 2026,

)

P 42803569

— bacteria, archaea, eukaryotes.

- **Orthologs / NCBI identifiers:** human *PLPBP* (Gene 65015); mouse *Plpbp* (NCBI Taxon 10090); zebrafish *plpbp* (Taxon 7955, ZFIN); *E. coli* **yggS** (Taxon 562); *Salmonella enterica* **yggS** (Taxon 28901); *S. cerevisiae* **YBL036C** (Taxon 4932); cyanobacterial **PipY**.
- **Natural disease in animals: no naturally occurring companion-animal/wildlife disease documented (OMIA: none established)** — human disease and engineered models only. **Veterinary relevance: minimal/not applicable.**
- **Comparative biology:** loss of the ortholog disrupts **amino-acid metabolism via PNP** across species (bacteria: Ito 2019/2020 PMIDs 30902856, 31677193; *Salmonella* aspartate metabolism, Vu & Downs 2021

),

P 34498310

indicating deep conservation of the **PLP-homeostasis mechanism**.

- **Transmission / zoonosis: not applicable** (genetic disease).

15. Model Organisms

- **Zebrafish (*Danio rerio*, Taxon 7955) — best vertebrate model.** CRISPR/Cas9 **plpbp−/−** larvae: behavioural, biochemical, and electrophysiological **seizures by 10 dpf**, **death by 16 dpf**, with **disrupted amino-acid, GABA and catecholamine metabolism**; **pyridoxine rescues** the phenotype and extends lifespan (Johnstone 2019,

).

P 30668673

Phenotype recapitulation: high (seizures, early lethality, B6-responsiveness). **Limitation:** does not capture human cortical white-matter/structural changes or long-term cognition. Resource: ZFIN.

- **Bacterial models — mechanistic workhorses.** *E. coli* **yggS** and *Salmonella* **yggS** mutants reveal **PNP accumulation**, glycine-cleavage-system inhibition and amino-acid metabolic disruption (Ito 2019/2020 PMIDs 30902856, 31677193; Vu & Downs 2021

).

P 34498310

Limitation: no nervous system; purely biochemical.

- **Yeast (*S. cerevisiae* YBL036C) and cyanobacterial PipY** — structure/function and PLP-independent/moonlighting roles (Llop 2026, P 41820517).
- **In vitro / structural:** bacterial YggS crystal structures (TIM-barrel; He 2022 P 35897955); human PLPHP biochemistry incl. **RNA-binding** (Graziani 2024 P 39604152); AlphaFold model (UniProt O94903).
- **Mammalian (mouse *Plpbp*):** ortholog exists and PLPBP biology probed indirectly (e.g., CLPP-KO mitochondrial study, Key 2023 P 38139332); **no widely-reported dedicated *Plpbp* mouse epilepsy model** (*gap / not available*). Resources: MGI, IMPC.

Summary of Supported vs. Open Questions

Supported (high confidence): - Monogenic AR disease from biallelic *PLPBP* LoF variants disrupting PLP homeostasis (P 27912044, P 30668673). - Neonatal-onset, AED-resistant, **B6-responsive** epileptic encephalopathy (P 41671913, P 36308585, P 33425341). - Mechanism: PLPHP chaperones/buffers PLP and prevents toxic PNP; loss impairs PLP-dependent amino-acid/neurotransmitter enzymes (P 35897955).

P 31677193).
P 34978460
- No specific biomarker → diagnosis by B6 trial + molecular testing ( ).
P 31741821
- Guarded prognosis despite treatment; genotype–severity correlation (
P 41671913
).
P 30668673
- Deep cross-species conservation; zebrafish/bacterial models; lysine-reduction therapy inapplicable (
P 42803569
P 30668673
).
P 36008148

Open / not available: - Precise prevalence, penetrance, carrier frequency, survival/QoL metrics. - Human omics atlases (transcriptomic/proteomic/metabolomic) of affected brain. - Dedicated mammalian (mouse) epilepsy model; approved advanced therapeutics (gene therapy). - Full in-vivo clarification of the GABA/glycine/catecholamine contributions and the moonlighting RNA-binding/mitochondrial roles in human pathogenesis.

Key references (PMID)

27912044 (Darin 2016, discovery) · 30668673 (Johnstone 2019, cohort + zebrafish) · 41671913 (Alsini 2026, systematic review n=54) · 34978460 & 31677193 & 30902856 (Ito, PNP mechanism) · 35897955 (He 2022, structure) · 39604152 (Graziani 2024, RNA-binding) · 38139332 (Key 2023, mitochondrial pool) · 42803569 (Xu 2026, conservation) · 34498310 (Vu & Downs 2021, Salmonella) · 41820517 (Llop 2026, PipY) · 31741821 (Jensen 2019, diagnostic pitfalls) · 33425341 (Ahmed 2020) · 36308585 (İpek 2022) · 36577449 (Kalser 2023, spasms) · 42563912 (Takle 2025, OMAS mimic) · 33977028 (Pal 2021, founder) · 28391250 (Plecko 2017) · 16159904 (Been 2005, epidemiology) · 36008148 (Coughlin 2022, LRT/ALDH7A1) · 40832147 (Coughlin & Gospe 2023, PDE review).